
WELCOME TO
ASTROJOURNEY
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OUTER
SPACE

IS HUGE!
LIKE REALLY.

THE MILKY WAY IS AT LEAST
100,000 LIGHT YEARS IN DIAMETER
THAT IS 946,073,047,258,004,200 KM
THAT IS 2,461,168,177,050 TIMES THE DISTANCE MOON - EARTH

NUMBERS SO HUGE, WE CANNOT UNDERSTAND.

OUR
GOAL

START SMALL - GO BIG - GET IMMERSED - HAVE FUN EXPLORING

MAKE THE SIZE OF MOONS, PLANETS, THE WHOLE SOLAR SYSTEM AND MUCH MORE ... UNDERSTANDABLE

MAKE THE ENORMOUS SIZE OF CELESTIAL BODIES AND DISTANCES BETWEEN THEM TRULY BE FELT

BUILD THE MILKY WAY FROM REAL DATA AND MAKE IT EXPLORABLE IN VR

THE CHALLENGE

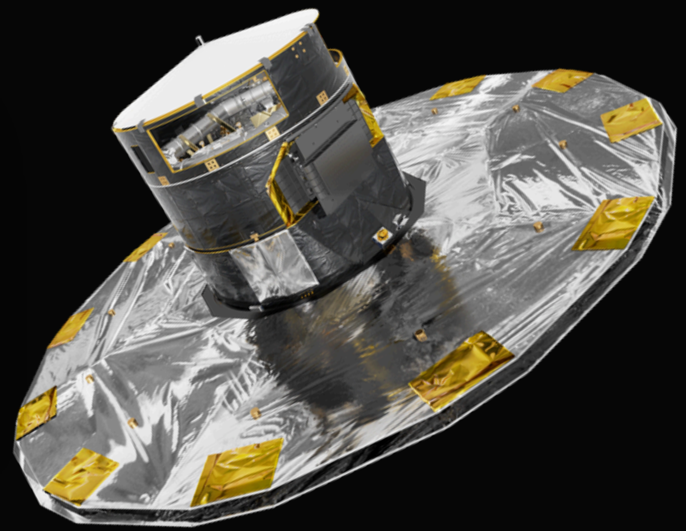
GAIA 3 DATA CONTAINS ABOUT 1.81 BILLION SOURCES (1.3 TB)

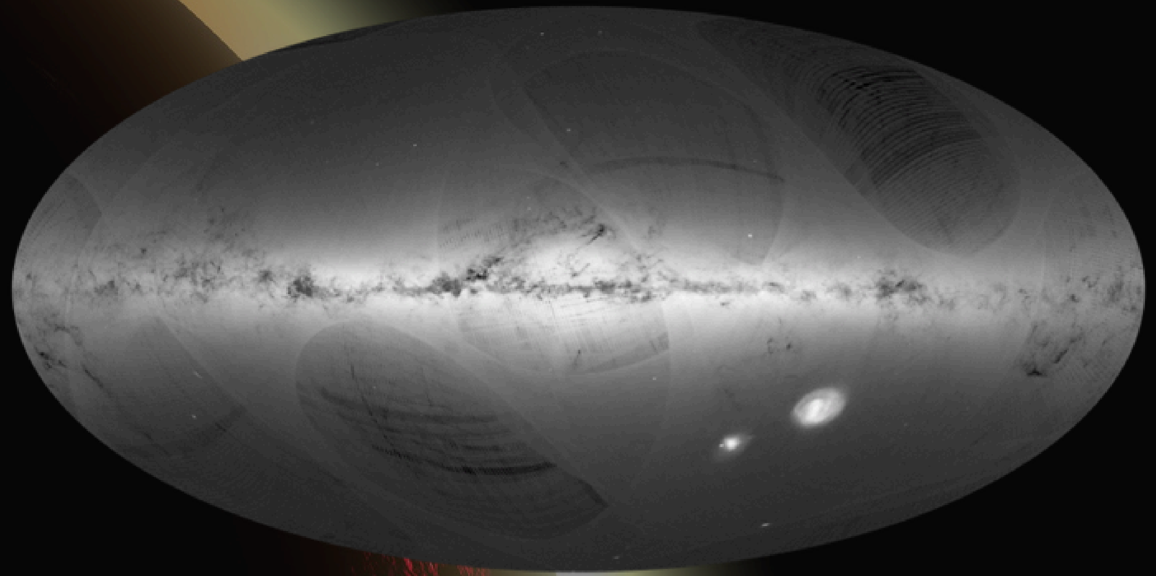
RENDERING 1810000000 STARS IS A LOT FOR OUR ENGINGE OF CHOICE: UNITY 6

HOW DO WE EVEN RENDER THE SHEER AMOUNT AND SPATIAL SIZE OF THE DATA?

HOW CAN WE NAVIGATE SPACE IN MINUTES INSTEAD OF LIFE TIMES?

THE DATA





WHERE DO WE GET THE DATA FROM?
- GAIA DR3 STAR CATALOG FILES FROM THE OFFICIAL ESA GAIA ARCHIVE

HOW DO WE HANDLE BIG DATA?
- 1.3 TB ZIPPED CSV FILES
- DOWNLOAD STEP BY STEP, FILTER ON THE GO, COMPRESS, OPTIMIZE

HOW IS THE DATA FILTERED?
- ONLY STARS WITHIN A CYLINDER AROUND THE MILKY WAY DISC
- KEEP N BRIGHTEST STARS BY ABSOLUTE MAGNITUDE (TOP-K SELECTION)
- KEEP ONLY ESSENTIAL PARAMETERS (RIGHT ASCENSION (RA), DECLINATION (DEC), DISTANCE (PC), ABSOLUTE MAGNITUDE, SOURCE ID)

MATH AND GEOMETRY

THE CORE IDEA: DEPICT INFINITY VIA PROJECTION

PARALLAX CALCULATION FOR PLANETS, ASTEROIDS AND STARS

SHOW CELESTIAL OBJECTS ON VIRTUAL SPHERICAL HORIZON AROUND THE PLAYER

MOVE THE DATA, NOT THE PLAYER

DISTANCES DIFFER BY WHOLE ORDERS OF MAGNITUDE

- FLOATING-POINT ERRORS
- HIERARCHICAL SECTOR BASED COORDINATE SYSTEM

OPTIMIZE
PERFORMANCE

THE CORE IDEA: DEPICT INFINITY VIA PROJECTION

PARALLAX CALCULATION FOR PLANETS, ASTEROIDS AND STARS

- SHOW CELESTIAL OBJECTS ON VIRTUAL SPHERICAL HORIZON AROUND THE PLAYER

MOVE THE DATA, NOT THE PLAYER

- PLAYER MOVES THROUGH DATA SPACE, NOT GAME SPACE

DISTANCES DIFFER BY WHOLE UNITS OF MAGNITUDE

- FLOATING-POINT ERRORS
- → HIERARCHICAL, SECTOR BASED COORDINATE SYSTEM

USING SHADERS

DON'T USE LIGHTS, FAKE THEM

- PLANET SHADOW SHADERS COMBINE UP TO 3 TEXTURES AND DRAW SHADOW ON THE SHADE SIDE

BURST COMPILER AND GPU COMPUTE SHADERS

- PARALLEL THREADING AS MUCH AS POSSIBLE

PERFORMANT QUAD MESHES AS STAR 'POINT SPRITES'

- MANY INSTANCES CAN BE DRAWN IN ONE CALL

OPTIMIZED AND FAST CULLING

- DOT PRODUCT FOV CULLING, STRIDE-BASED SAMPLING

ASTROJOURNEY

Andreas Schiefner

It's a great view from here. I'm having the time of my life. - Michael P. Anderson

Philipp Unger

The stars don't look bigger, but they do look brighter. - Sally Ride

GET PROJECT: HTTPS://GITHUB.COM/GRAFKNUSPRIG/PARALLAX_SIM